

PSYCHOLOGY

People Differ Widely in Their Understanding of Even a Simple Concept Such as the Word ‘Penguin’

We think about what a penguin is like in dozens of different ways—one reason why we often talk past each other

By Simon Makin on April 25, 2023



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The Merriam-Webster dictionary defines the word “penguin” as “any of various erect short-legged flightless aquatic birds (family Spheniscidae) of the southern hemisphere.” That description seems simple enough, but definitions are not what people have in mind when they actually use words. Instead people think of concepts: the myriad properties, ideas, examples and associations that spring to mind when we think about a word.

Our concepts are crucial to exactly what we mean when we use language, and new research has found that the concepts people hold, even for a word like penguin, vary from person to person on a shockingly frequent basis. This does not mean we all disagree on the basic definition of a penguin. But while some people might think they are noisy, plump creatures, more like a whale than an eagle, others might consider them to be awkward, strange animals, more like an ostrich than a dolphin.

These discrepant views—these concepts of penguins—are the kind of information researchers at the University of California, Berkeley, elicited from participants in a study that was published last month. The team’s results show that even the plainest of nouns can invoke dozens of distinct concepts in individuals’ mind. “People have wondered for a long time how to put a number on how much overlap there is, and it’s really low. It blows my mind,” says psychologist Celeste Kidd of the U.C. Berkeley, who was senior author of the study.

To make matters worse, the researchers found that people are usually oblivious to these differences and believe that most other people think like they do even when almost nobody does. This may be one reason people so often are at loggerheads. “We think it can explain a lot of disagreements people have,” Kidd says. “It’s an approach to understanding why people talk past each other.” Being more aware of how often we might not be comprehending one another may help us “get on the same page when it matters,” she adds.

Determining the extent to which people’s concepts about things match up is a problem that has long occupied philosophers, psychologists and linguists alike. It is well known that abstract, high-concept words such as “knowledge” and “fairness” provoke frequent debates about exactly what is meant. But researchers have struggled to formally characterize how people’s concepts differ and to quantify how often that happens. Past efforts have stumbled, Kidd says, because we do not fully understand what concepts consist of.

That is not to say researchers have no idea. A famous study in 1984 found that the concepts young children have for words often consist of collections of observable properties, known as features. Take the word “uncle,” for example. “For a four-year-old, an uncle is someone who gives you presents at Christmas and is pally with your parents,” says psychologist James Hampton of City, University of London, who was not involved in Kidd’s study. Older children develop concepts that have some reasoning behind them. “For an eight-year-old, an uncle is a male sibling of a parent,” Hampton says. “They shift from something based on the features they observe to understanding something deeper, more relational.” This more sophisticated understanding that evolves as children interact with their surroundings is described by a model of how cognition develops that psychologists call “theory theory.”

Hampton’s work has promoted the idea that “prototypes” are the basis of concepts. Prototypes are sets of features that determine how typical a specific example of something is in terms of a broader category. A blackbird is closer to the prototypical bird than a penguin is, for example. There is a large body of work behind these ideas, and they likely all play a role in human cognition. But that still leaves the much debated question of what exactly is in

people's mind when they contemplate any given word. And how similar people are in this respect has been a mystery until now.

Instead of trying to address debates about the nature of concepts, Kidd and her team sidestepped them using methods that try to detect *some* but not all differences among people. In the researchers' first experiment, they asked nearly 1,800 participants for similarity judgments, such as "What's most similar to a penguin, a finch or a dolphin?" In the second, they asked for feature judgments such as "Is a penguin noisy?" They then used mathematical clustering methods to estimate the number of distinct concepts that existed in the sample of participants and subsequently to extrapolate that estimate to the world at large. Surprisingly, the team's results suggest that at least 10 to 30 quantifiably different concept variants exist for even common nouns such as penguin. These are rough but probably conservative estimates, the researchers say.

The participants did not just disagree about penguins. The researchers found different concepts existed for every word they used, with people disagreeing on important matters such as whether seals are graceful. In addition to animals, the team used politicians' names, such as George W. Bush and Joe Biden, and found even more variation in participants' concepts for these words. "This backs up what lots of people have been nibbling away at for a while with actual data," says neuroscientist Kris De Meyer of King's College London, who was not involved in the study but works on the communication of climate change.

The researchers expected that people's concepts about politicians would vary widely because of different political beliefs, but the amount of variation they saw for basic animal concepts surprised them. "The probability two people selected at random will share the same concept about penguins is around 12 percent," Kidd says. For example, "people disagree about things like whether penguins are heavy, presumably because they haven't lifted a penguin." Here, Kidd hints at where a lot of these differences likely come from: they boil down to a person's life experiences "If you've spent time watching penguins walk, you're maybe more likely to think they're heavy from their waddle," Kidd says. "If you've spent time learning about anatomy, you've maybe learned birds have light skeletons and so think penguins are light." Those differences manifest themselves in other ways. Professional philosophers have a very different concept of "knowledge" than most people. Researchers have studied this effect of training, but such observations tell us little about how often people's concepts differ. Kidd and her colleagues' findings regarding simple nouns such as penguin suggest that conceptual differences are so common that they are probably fundamental to how we think about things.

The study does not show that people have no understanding in common—no one confuses penguins with albatrosses—just that we may have less common ground than previously appreciated. "There's something that's shared," says psychologist Susan Gelman of the University of Michigan, who studies the relationship between language and thought and was not involved in Kidd's study. "But maybe everything that gets attached to [words] is a lot more idiosyncratic and varied than we thought."

In the recent study, the researchers also asked participants to say what proportion of people would agree with their similarity judgments and compared these responses with the actual numbers. Participants believed around two thirds would agree with them when the actual proportion was usually much smaller. In some cases, people believed they were in the majority when virtually nobody agreed. This shows that people are typically oblivious to the extent to which others share their concepts. “That was cool,” Gelman says, “and may have implications for when we think we’re communicating but aren’t.”

The findings could be relevant to disagreements about more serious matters than whether a penguin is heavy. “If this happens with common nouns, how much worse can it be with abstract words we use to describe the big problems we’re dealing with?” De Meyer says. “These methods would show people have different conceptualizations of words like inequality, fairness, climate change, two degrees and everything else we use to describe what’s important to us.”

The study may offer hope that we can sometimes overcome our differences by simply becoming aware of them. “When people disagree, it might not be for the reasons they think,” Kidd says. “It could just be because their concepts aren’t aligned.” Her advice: “Hash it out,” she says. “Questions like ‘What do you mean?’ can go a long way toward preventing a dispute going off the rails.”

ABOUT THE AUTHOR(S)



Simon Makin is a freelance science journalist based in the U.K. His work has appeared in *New Scientist*, the *Economist*, *Scientific American* and *Nature*, among others. He covers the life sciences and specializes in neuroscience, psychology and mental health. Follow Makin on Twitter [@SimonMakin](https://twitter.com/SimonMakin).

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